

Assignment 15

1. Consider a nuclear-heated rocket, operating in vacuum ($p_a = 0$). The propellant, hydrogen (H_2) enters the heat exchanger at stagnation temperature of 70 K and stagnation pressure of 2 MPa. The heat exchanger has friction coefficient of 0.005, and length-to-diameter ratio (L/D) of 600. The maximum allowable wall temperature ($T_{w,max}$) is 2800 K. The nozzle contraction ratio (A_c/A_t) is 1.25 and the exit-to-throat area (A_e/A_t) is 10. $\gamma = 1.3$.
 - a) Calculate the stagnation temperature at the exit of the heat exchanger. {Ans.: 2410 K}
 - b) Find the Mach number at the heat exchanger entry. {Ans.: 0.037}
 - c) What is the stagnation pressure loss in the heat exchanger? {Ans.: 53.1%}
 - d) What is the rocket specific impulse? {Ans.: 817 s}
2. Is the maximum temperature possible in a nuclear-heated rocket greater or smaller than that in a chemical rocket? Explain.
3. What is the primary reason for the higher specific impulse provided by a nuclear-heated rocket (compared to its chemical counterpart)?
4. Describe the features of the heat exchanger used in the NTR of the NERVA program.